

University of Warsaw
Faculty of Mathematics, Informatics and Mechanics

Ignacy Makowski

Student no. 418408

Proteomic profiling of *S. cerevisiae*
strains perturbed in RNA polymerase
III

Master's thesis
in BIOINFORMATICS AND SYSTEMS BIOLOGY

Supervisor:
prof. dr hab. Anna Gambin
Institute of Informatics

Warsaw, October 2025

Abstract

This thesis aims to expand upon previous findings that glycolytic flux in *Saccharomyces cerevisiae* is dependent on RNA Polymerase III (RNAP III) and its negative regulator Maf1. The primary objective was to comprehensively understand how the absence of MAF1 in the *maf1* Δ strain and a point mutation in the RNAP III subunit in the *rpc128-1007* strain influence mitochondrial fitness at the proteome level under both fermentative and non-fermentative growth conditions [12]. To achieve a broader comparative context, the study was expanded by incorporating reanalyzed proteomic data from the common laboratory strains W303 and BY4742 [11] alongside newly generated data from the *maf1* Δ and *rpc128-1007* mutant strains. In both the new and existing datasets, protein levels were measured using mass spectrometry. The ComBat algorithm was used to correct for inter-experimental batch effects, which allowed reliable combination of data from multiple sources [4]. The integrated data were subsequently analyzed using a suite of statistical methods, including generation of heatmaps for hierarchical clustering, volcano plots to identify differentially abundant proteins, and box plots to visualize data distribution before and after correction. Mapping protein expression changes onto the Oxidative Phosphorylation pathway [6] revealed altered abundance of key complex components. An additional investigation into a potential link between protein abundance and codon usage (AAA, CAA, GAA, AAG) was performed. Scatter plots comparing protein level changes against the frequency of target codons showed no significant correlation. In conclusion, this integrated proteomic approach demonstrates how genetic alterations in the RNAP III machinery impact mitochondrial fitness and protein abundance across different growth conditions. By harmonizing new and existing datasets, the study provides a comprehensive

map of the metabolic shifts occurring in *Saccharomyces cerevisiae* mutants, beyond simple codon-usage dependencies.

Keywords

Proteomics, Mass spectrometry, RNA Polymerase III, Oxidative phosphorylation, *maf1Δ*, *rpc128-1007*

Thesis domain (Socrates-Erasmus subject area codes)

11.2 Statistics

Subject classification

62. Statistics

92. Biology and other natural sciences

Tytuł pracy w języku polskim

Profilowanie proteomiczne szczepów *S. cerevisiae* z zaburzoną aktywnością polimerazy RNA
III

Contents

1. Introduction	5
2. Background	7
2.1. Theoretical background	7
2.1.1. RNA polymerases	7
2.1.2. Structural Differences Between Human and Yeast RNAP III	7
2.1.3. <i>Saccharomyces cerevisiae</i> <i>maf1</i> Δ and <i>rpc128-1007</i> mutations	7
2.1.4. Laboratory Yeast Strains: W303 and BY4742	9
3. Materials and Methods	11
3.1. Overview of the Study Design	11
3.2. Computational Methods	11
3.3. Data Sources and Initial Processing	12
3.4. Integrative Analysis and Batch Correction	12
3.5. Statistical Analysis and Visualization	12
3.6. Codon Usage Analysis	13
4. Results and discussion	15
4.1. Proteomic Profiling of RNAP III Mutants	15
4.2. Coefficient of Variation Analysis	17
4.3. Differential Abundance	17
4.4. Codons	18
4.5. KEGG Pathway Enrichment Analysis	20
4.6. KEGG	23
4.7. Functional Analysis	23
4.8. Integrative Analysis of Mutant and Laboratory Strains	26
5. Additional Materials	29

Chapter 1

Introduction

The regulation of metabolism in *Saccharomyces cerevisiae* has been extensively studied, revealing significant insights into yeast's proteome homeostasis. Among the key components contributing to it is RNA Polymerase III (RNAP III), whose activity in producing non-coding RNAs is essential for sustaining cellular growth [12]. Using *maf1Δ* and *rpc128-1007* yeast mutants with altered tRNA abundance, it was demonstrated that disturbed RNAP III activity leads to major shifts in TCA cycle metabolism. ^{13}C flux analysis and TCA enzyme activity profiling revealed metabolic constraints, including reduced flux toward glutamate, aspartate, and phosphoenolpyruvate, explaining impaired growth in certain conditions. Notably, *rpc128-1007* exhibited increased TCA cycle activity under non-fermentable conditions. These results indicate that balanced non-coding RNA synthesis, tightly linked to glucose signaling, is essential for mitochondrial metabolic flexibility and maintaining cellular homeostasis [12]. Prior work has established that dysregulation of this pathway has significant metabolic consequences, directly linking RNAP III activity to glycolytic flux and mitochondrial function [12]. However, the global proteomic shifts that underpin these metabolic phenotypes remain incompletely characterized. This study was designed to comprehensively map the proteomic landscape shaped by altered RNAP III activity. We utilized two key mutant strains: *maf1Δ*, which lacks the Maf1 repressor and thus exhibits constitutively high RNAP III activity, and *rpc128-1007*, which carries a mutation in an RNAP III subunit related to assembly defects [10]. The *maf1Δ* strain displays severe growth defects on non-fermentable carbon sources; this phenotype is reversed by the *rpc128-1007* mutation, providing a tractable model to investigate the consequences of increased and decreased tRNA synthesis [12]. The primary objective was to use SWATH mass spectrometry to quantify proteome-level changes in these strains under both fermentative (glucose) and non-fermentative (glycerol) conditions, thereby elucidating the molecular basis of their distinct mitochondrial fitness. We integrated our newly generated data with a public proteomic dataset from common laboratory strains W303 and BY4742 [11].

Chapter 2

Background

2.1. Theoretical background

2.1.1. RNA polymerases

Prokaryotic organisms possess a single RNA polymerase. In contrast, the RNA synthesis process in eukaryotic organisms is more complex, as it is carried out by three distinct enzymes: RNA polymerases I, II, and III. Polymerases I and II synthesize rRNA and mRNA, which are essential for ribosome assembly and protein synthesis. RNA polymerase III is responsible for transcribing 5S rRNA, tRNAs, U6 snRNA, and other small non-coding RNAs. These molecules play key roles in translation, RNA processing, and the regulation of cellular functions. This polymerase is particularly important because its dysfunction is linked to severe diseases, such as neurodegenerative disorders and cancers. [13]

2.1.2. Structural Differences Between Human and Yeast RNAP III

While the overall architecture and catalytic core of RNA Polymerase III (RNAP III) are highly conserved across eukaryotes, notable structural differences exist between the human and yeast enzymes, primarily within their peripheral subcomplexes [13]. For instance, the human RPC10 subunit contains a zinc ribbon domain that remains near the active center during transcription elongation, whereas the corresponding yeast C11 domain completely dissociates from the enzyme core [13]. Additionally, human RNAP III utilizes a specific iron-sulfur cluster in the RPC6 subunit to stabilize its initiation complex, a motif that is replaced by a simple structural loop in the yeast C34 homolog [13]. The human RPC5 subunit also features a unique extended tail with winged-helix domains that are crucial for maintaining the overall stability of the complex [13]. Finally, the human enzyme employs a structural autoinhibitory mechanism where the RPC7 subunit blocks the DNA-binding cleft to prevent non-specific transcription, a regulatory feature that has not been observed in the single yeast equivalent, C31 [13].

2.1.3. *Saccharomyces cerevisiae* *maf1* Δ and *rpc128-1007* mutations

The *maf1* Δ mutant strain is characterized by the deletion of the *MAF1* gene, which encodes the negative regulator of RNA Polymerase III (RNAP III) in *Saccharomyces cerevisiae* [12]. Consequently, this deletion leads to constitutively high RNAP III activity and elevated levels of tRNAs that are decoupled from environmental signaling [12]. Metabolically, *maf1* Δ cells undergo a reprogramming similar to the Warburg effect, shifting towards accelerated glycoly-

Table 2.1: Subunits of Yeast and Human RNA Polymerase III [13]

Domain / Subcomplex	Yeast (Pol III)	Human (Pol III)
Polymerase Core	C160	RPC1
	C128	RPC2
	AC40	RPAC1
	AC19	RPAC2
	Rpb5	RPABC1
	Rpb6	RPABC2
	Rpb8	RPABC3
	Rpb10	RPABC4
	Rpb12	RPABC5
	C11	RPC10
	N-Zn-ribbon	N-Zn-ribbon
TFIIS	C11	RPC10
C-Zn-ribbon like	C-Zn-ribbon	C-Zn-ribbon
Polymerase Stalk	C17	RPC9
	C25	RPC8
TFIIF-related	C53	RPC4
	C37	RPC5
TFIIE-related	C82	RPC3
	C34	RPC6
	C31	RPC7

sis and redirecting glucose flux into the pentose phosphate pathway, as well as into glycogen and trehalose synthesis [12]. This mutant exhibits decreased tricarboxylic acid (TCA) cycle activity during fermentative growth and accumulates high levels of acetyl-CoA and lipids [12]. Furthermore, the *maf1Δ* strain displays severe growth defects, including lethality when cultured on non-fermentable carbon sources (such as glycerol) at elevated temperatures like 37°C [12].

In contrast, the *rpc128-1007* strain carries a point mutation in the C128 subunit of RNAP III, which impairs the assembly of the polymerase complex and reduces overall tRNA synthesis [10]. The defective assembly of the RNAP III complex halts cell division, resulting in an accumulation of unbudded cells arrested in the G1 phase of the cell cycle [10]. Metabolically, the *rpc128-1007* mutation severely limits the cell’s ability to utilize glucose through glycolysis [12]. To survive on glucose, this mutant relies on an alternative metabolic strategy driven by the Idp-dependent reductive carboxylation of glutamine, which helps maintain TCA cycle cataplerosis and supports lipid biosynthesis despite the glycolytic block [12]. When grown on non-fermentable carbon sources, however, the *rpc128-1007* strain adapts by maintaining a highly active TCA cycle [12].

Importantly, the *rpc128-1007* mutation acts as a phenotypic suppressor of the *maf1Δ* strain [12]. In a double mutant model, the reduced RNAP III activity caused by the C128 mutation rescues the severe growth defect of *maf1Δ* on non-fermentable carbon sources at restrictive temperatures [12]. While restoring tRNA synthesis levels in the double mutant does not rescue the cell cycle arrest caused by the primary assembly defect [10], it effectively rewires carbon flux to increase TCA cycle activity and restore mitochondrial metabolic flexibility [12].

2.1.4. Laboratory Yeast Strains: W303 and BY4742

The W303 and BY4742 strains are widely used *Saccharomyces cerevisiae* laboratory models with distinct genetic backgrounds and auxotrophic markers [11]. The W303 strain is of the MATa mating type and contains specific point mutations for auxotrophy, such as *trp1-1*, *ade2-1*, *ura3-1*, *his3-11,15*, and *leu2-3112* [11]. Researchers frequently choose this background for cell cycle studies because the cells are easy to synchronize [11]. In contrast, the BY4742 strain is of the MAT α mating type and features complete gene deletions for markers including *HIS3*, *URA3*, *LYS2*, and *LEU2* [11]. This strain is highly popular due to the commercial availability of numerous compatible genomic tools [11].

Chapter 3

Materials and Methods

3.1. Overview of the Study Design

The workflow of this study integrates experimental data generation with a multi-step computational pipeline. The methodology is structurally divided into four primary stages. First, the data sources are compiled by combining newly generated SWATH-MS proteomic profiles for the wild-type, *maf1Δ*, and *rpc128-1007* strains with a previously published dataset covering the common laboratory strains W303 and BY4742. Second, the datasets undergo integration and harmonization, which involves gene ID mapping, replicate averaging, log2 transformation, and batch effect correction using the ComBat algorithm to eliminate non-biological experimental variation. The third stage encompasses multivariate and statistical analyses, where dimensionality reduction techniques such as PCA and UMAP [8] are applied alongside differential abundance evaluation and codon usage correlation. Finally, the processed data is subjected to functional discovery, focusing on stress response mechanisms and mapping to the Kyoto Encyclopedia of Genes and Genomes (KEGG) reference pathways [6, 5] to uncover biological insights into oxidative phosphorylation and mitochondrial fitness.

3.2. Computational Methods

- **Coefficient of variation (CV)** will be defined as population or sample standard deviation divided by its mean and multiplied by 100 to get percentage value $CV = \frac{\sigma}{\mu} \cdot 100$ [1]
- **Sequential window acquisition of all theoretical mass spectra (SWATH-MS)** is a mass spectrometry data-independent acquisition strategy which splits the m/z range into windows. Ions in each window are then fragmented, and the spectra are recorded. The advantage of this approach is the detection of low-abundance proteins 5.1. [9, 3]
- **Principal Component Analysis (PCA)** is a statistical technique that reduces number of variables into uncorrelated principal components. It consist of following steps [2]:
 - subtract the mean from each of the data dimensions
 - calculate the covariance matrix of the centered data
 - get eigenvectors and the eigenvalues from the covariance matrix (representing the amount of variance)
 - sort eigenvectors by their eigenvalues in descending order

form a feature vector by taking the eigenvectors

$$PCA_{data} = (ev_1, \dots, ev_n)^T \times OriginalMeanSubtractedData^T$$

- **Uniform Manifold Approximation and Projection (UMAP)** is a dimension reduction algorithm published in 2018. [8]
- **Kyoto Encyclopedia of Genes and Genomes (KEGG)** is a database located at <http://www.genome.ad.jp/kegg/> storing gene functions, higher order functional information and chemical compounds. Additionally visualization tools are also provided. [6]
- **ComBat algorithm** is a statistical method designed to correct for non-biological experimental variation, commonly referred to as batch effects, that occurs across multiple batches of experiments [4]. Adjusting for these batch effects is crucial when combining multiple datasets, as failure to do so makes data integration difficult and can mask true biological phenomena [4]. The ComBat algorithm utilizes parametric and non-parametric empirical Bayes frameworks to filter out this technical noise [4].

3.3. Data Sources and Initial Processing

The analysis integrated two distinct proteomic datasets. The first was a re-analysis of published data from Rossio et al. (2023), containing protein quantifications for *S. cerevisiae* strains W303 and BY4742 in exponential and stationary growth phases [11]. The second dataset was newly generated for this study, containing SWATH-MS quantifications for wild-type, *maf1Δ*, and *rpc128-1007* strains grown under various conditions (YPD, YPGly 30°C, YPGly 37°C). Initial data processing was performed using an R script. Data was loaded from the respective source files (xlsx and csv). Protein identifiers from the new dataset were mapped to standard gene symbols using a provided mapping file to ensure consistency between the datasets. For both datasets, replicate measurements for each condition were averaged to obtain a single representative intensity value per protein per condition.

3.4. Integrative Analysis and Batch Correction

The two datasets were merged based on common gene symbols. This combined dataset was then log2-transformed to stabilize variance. A significant challenge in this integrative approach was the presence of strong batch effects, where technical variation between the two experiments masked the underlying biological signals. To address this, the ComBat algorithm from the *sva* R package was applied to the log-transformed data [4]. This method adjusts the data to remove non-biological experimental variation, allowing for a more accurate comparison across all samples. The effect of this correction was visualized using boxplots of protein intensity distributions and Principal Component Analysis (PCA) before and after ComBat application.

3.5. Statistical Analysis and Visualization

All statistical analyses and visualizations were performed in R.

- **Hierarchical Clustering and Heatmaps:** To visualize global protein expression patterns, heatmaps were generated using the *pheatmap* package. The data was Z-score scaled across proteins to highlight relative expression changes.

- **Principal Component Analysis (PCA) and UMAP:** PCA and Uniform Manifold Approximation and Projection (UMAP) were used for dimensionality reduction to visualize the major sources of variation in the data and to assess sample clustering.
- **Differential Abundance Analysis:** To identify proteins with statistically significant changes in abundance between conditions, two methods were employed. Z-test and limma package were used to compare conditions within the new dataset. For both methods, p-values were adjusted for multiple testing using the Benjamini-Hochberg method. Results were visualized using volcano plots.

3.6. Codon Usage Analysis

A separate analysis was conducted using a Python script to investigate a potential link between protein abundance changes and codon usage. Differential expression results were merged (from the Z-test analysis) with the frequency of specific codons (AAA, CAA, GAA, AAG) for each gene. It then categorized genes based on two criteria: whether their protein product was significantly increased, decreased, or unchanged, and whether the frequency of the target codons was high ($\geq 10\%$) or low ($< 10\%$). The number of genes in each combined category was counted and visualized as a heatmap.

Chapter 4

Results and discussion

4.1. Proteomic Profiling of RNAP III Mutants

We focused on the SWATH-MS dataset comprising the wild-type, *maf1* Δ , and *rpc128-1007* strains. Unsupervised clustering via Principal Component Analysis (PCA) and heatmap analysis (Figure 4.1) revealed that the primary source of proteomic variation is driven by the experimental conditions. The first major division separates samples based on the carbon source, isolating those grown on fermentative glucose (YPD) from those on respiratory glycerol (YPGly). Subsequently, the clustering is determined by the experimental temperature. Finally, at the lowest hierarchical level, the samples group according to the specific yeast genotype. This pattern indicates a transcriptional and translational reprogramming in response to the metabolic shift from fermentation to respiration [11]. Furthermore, these observations show that further downstream analyses are necessary to identify the key differentially abundant proteins that explain the distinct phenotypic responses of the mutant strains [12].

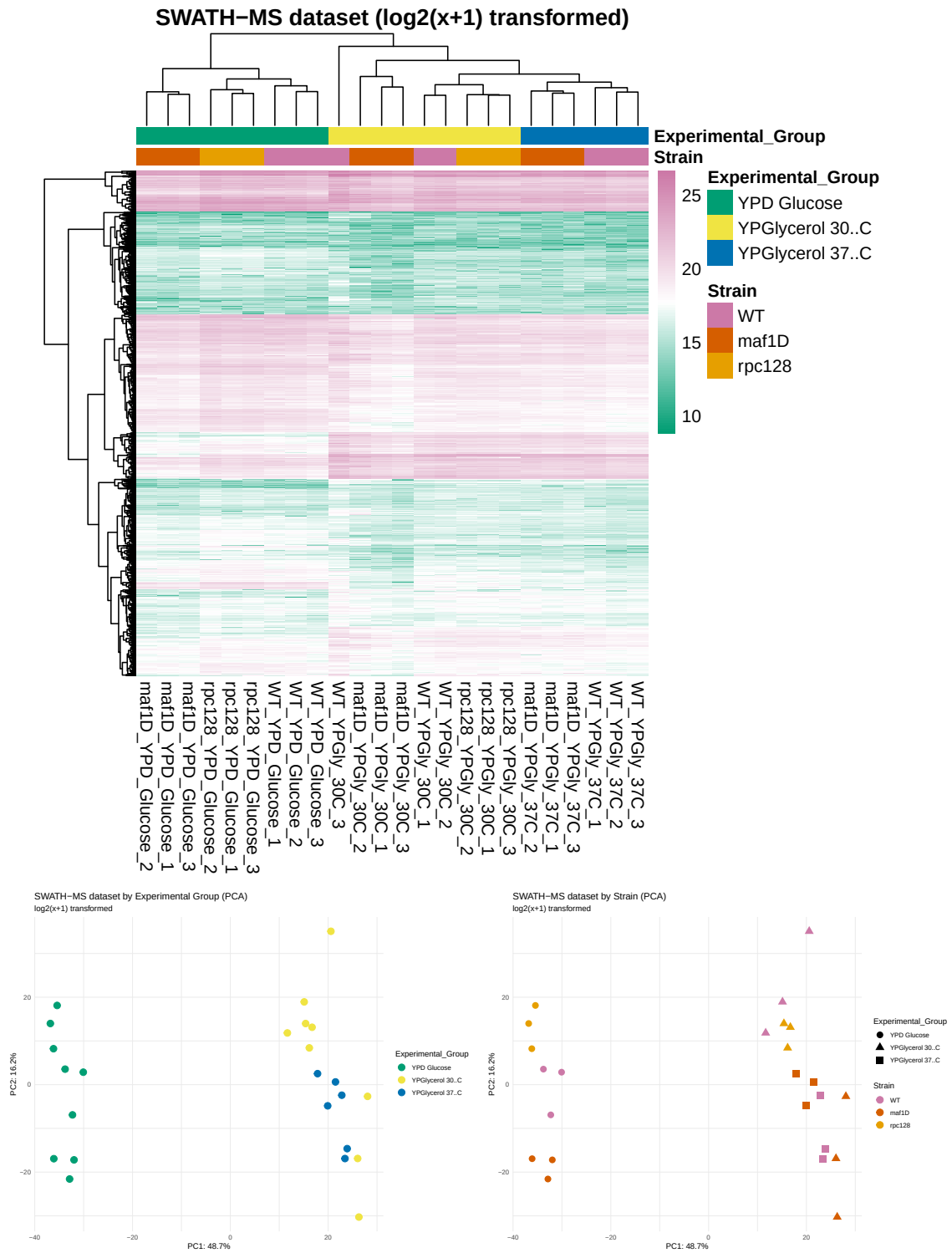


Figure 4.1: Global Proteomic Analysis of RNAP III Mutants. (Top) The heatmap shows hierarchical clustering of proteins and samples, clearly separating experimental conditions. (Bottom) Principal Component Analysis (PCA) visually confirms these clusters. Both analyses demonstrate that samples cluster primarily by carbon source, then by temperature and strain.

4.2. Coefficient of Variation Analysis

To assess the reproducibility and precision of protein quantification in the analyzed samples, we calculated the coefficient of variation (CV) distributions. Figure 4.2 presents box plots depicting the spread of CV values for individual genotypes (WT, *maf1* Δ , *rpc128-1007*) depending on the carbon source (YPD, YPGly) and temperature (30°C, 37°C).

The obtained median CV values range from 12.4% to 27.9%. This level of within-group variance is acceptable for large-scale label-free profiling measurements. The SWATH-MS technique provides high consistency and accuracy of protein quantification across multiple samples [3]. The limit of quantification for this method is defined as a deviation from measurement linearity exceeding 30% [3]. The presented distributions, with medians below this threshold, suggest that technical and biological variability within replicates remains at an acceptable level.

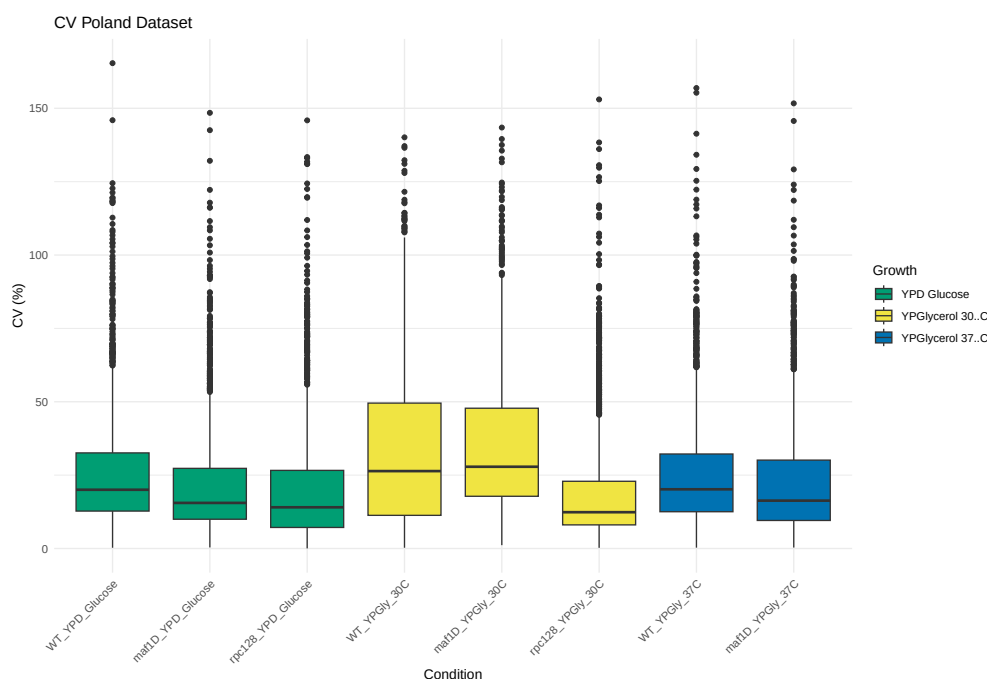


Figure 4.2: Coefficient of variation (CV) distributions across experimental conditions and genotypes. These boxplots show the spread of protein abundance variability.

4.3. Differential Abundance

We performed differential abundance analysis using the limma package. The most visible proteomic shift was observed when comparing the *maf1* Δ mutant to wild-type under respiratory conditions (YPGly 30°C). As shown in the volcano plot (Figure 4.3A), a large number of proteins were significantly upregulated in the mutant, indicative of a severe cellular stress response. A similar, though less pronounced, stress response was also seen at 37°C (Figure 4.3E). Other comparisons, such as between the mutants and wild-type on glucose (YPD), showed more subtle changes (Figure 4.3B, D).

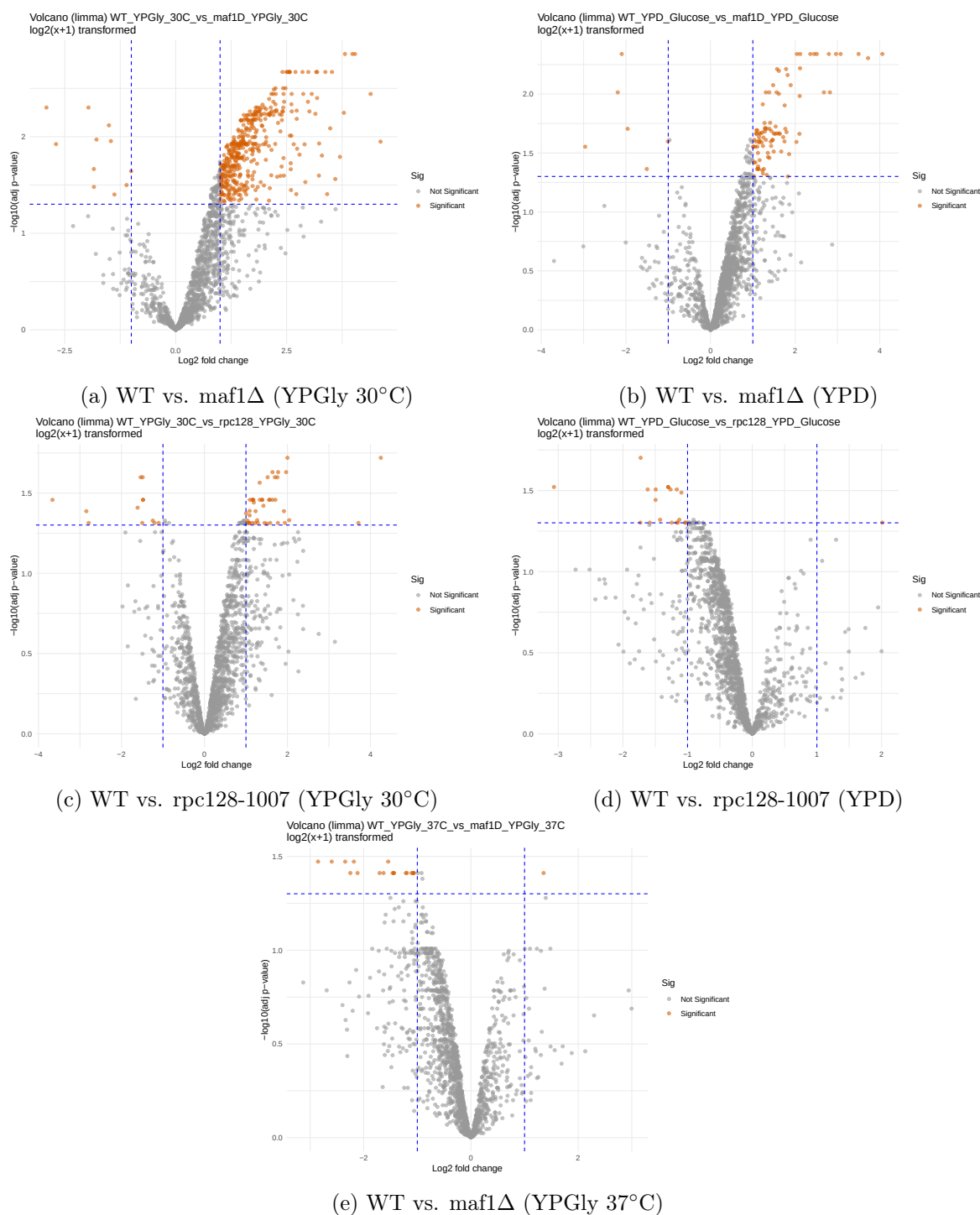


Figure 4.3: Volcano Plots of Differential Protein Abundance based on limma analysis.

4.4. Codons

Scatter plots of protein Z-scores versus codon frequency showed no correlation (Figure 4.4). A categorical analysis further supported this finding (Figure 4.5), indicating that the frequency of these specific codons is not a primary driver of the observed proteomic changes.

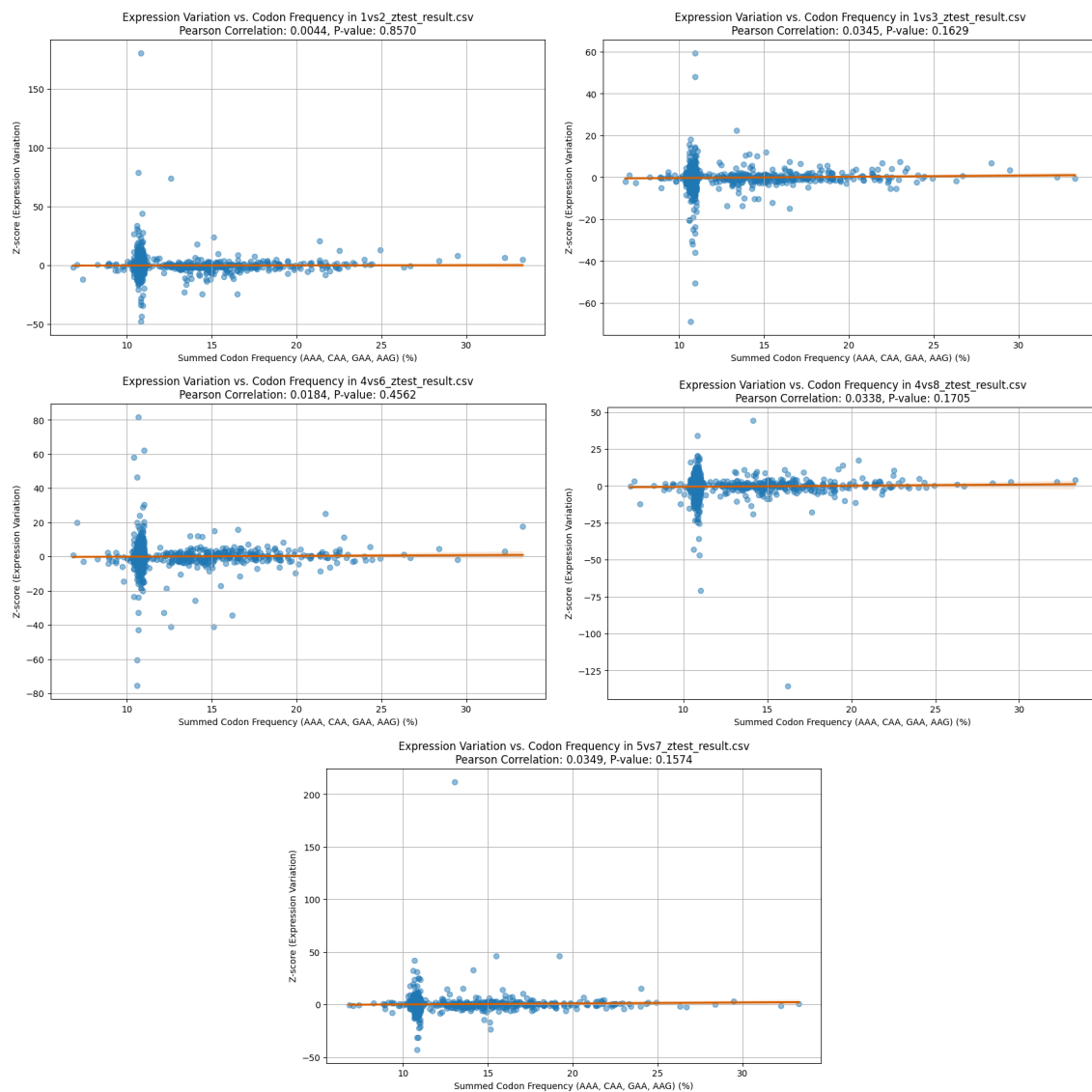


Figure 4.4: Protein Expression vs. Codon Frequency. Scatter plots showing the lack of correlation between protein Z-scores and the frequency of target codons for two representative comparisons.

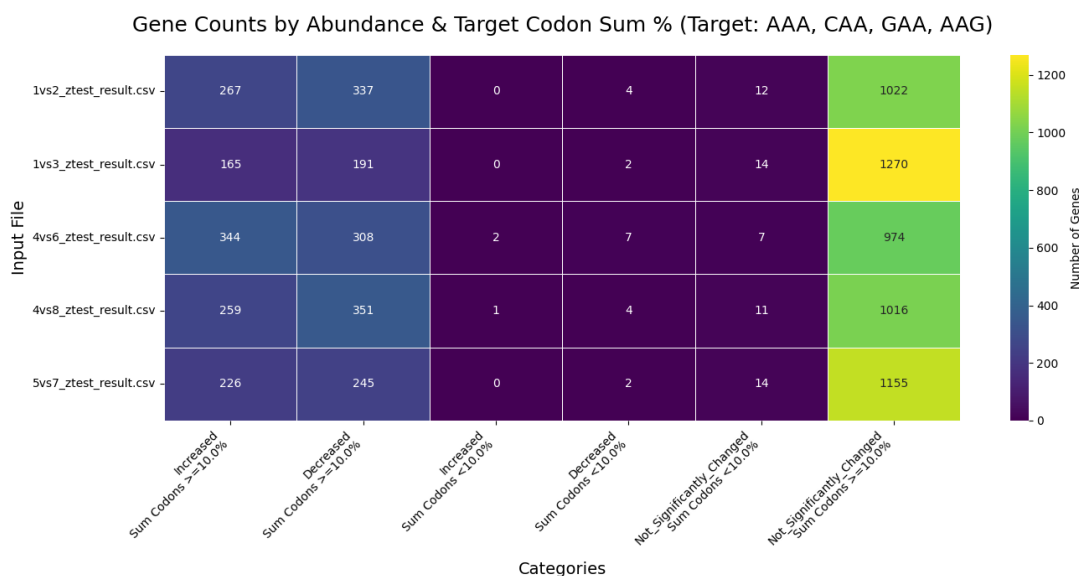


Figure 4.5: Categorical Analysis of Gene Counts. This heatmap shows the number of genes in categories defined by both abundance change and codon frequency.

4.5. KEGG Pathway Enrichment Analysis

Differentially abundant proteins were mapped to KEGG pathways. Pathway enrichment was evaluated using a 10 percent overlap threshold to identify affected metabolic networks. During growth on glucose, the *maf1* Δ strain compared to wild type showed among others an increased abundance of proteins related to the ribosome, oxidative phosphorylation, protein processing in endoplasmic reticulum pathways (Figure 4.6).

In contrast, the *rpc128-1007* strain exhibited only a decreased abundance in pathways with similar overlap. Glyoxylate and dicarboxylate metabolism, ribosome biogenesis in eukaryotes, purine metabolism, pyruvate metabolism and alanine, aspartate and glutamate metabolism shows gene count of 10 and over (Figure 4.7).

From the enriched KEGG terms, the pyruvate metabolism appear in both experiments Therefore, this is a candidate for explaining the differences between mutants 4.6, 4.7.

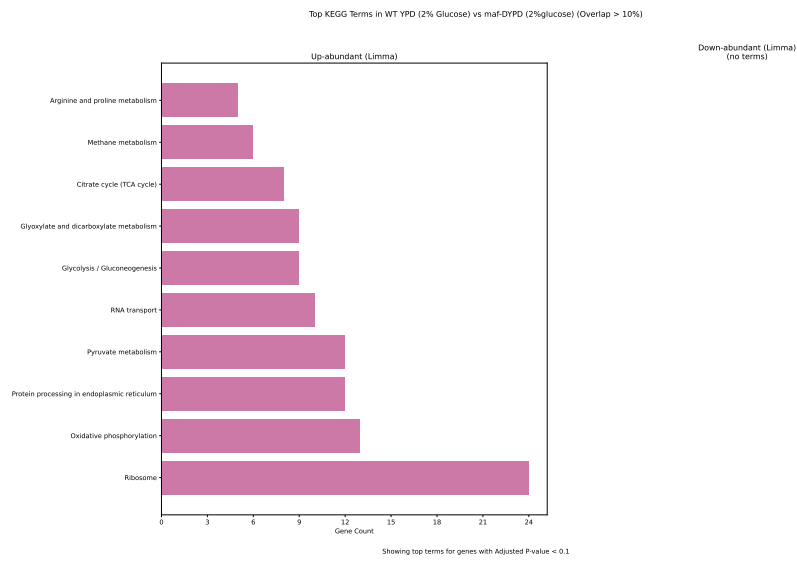


Figure 4.6: KEGG pathway enrichment for wild-type versus *maf1*Δ strains grown on glucose (YPD).

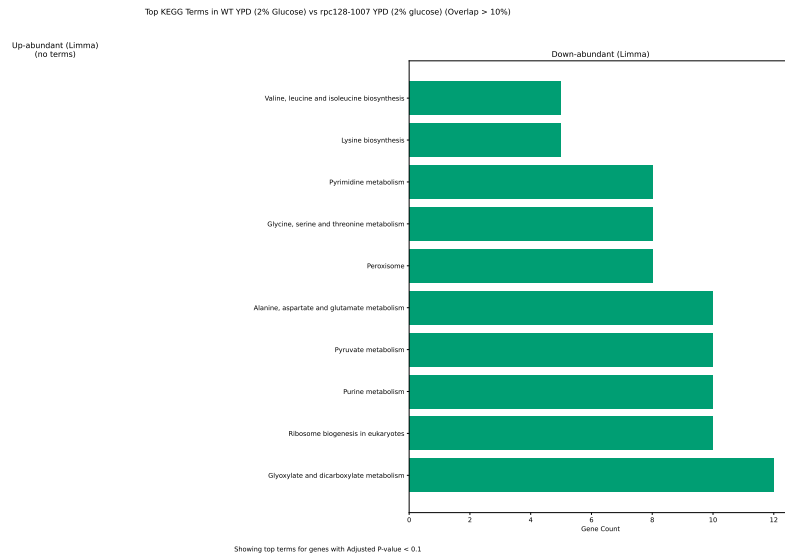


Figure 4.7: KEGG pathway enrichment for wild-type versus *rpc128-1007* strains grown on glucose (YPD).

On glycerol base, compared to the wild type strain, both mutants show increased abundance in terms Glycolysis / Gluconeogenesis, Glycine, serine and threonine metabolism, Cysteine and methionine metabolism, Methane metabolism so in both cases is above 40% of all increased paths. In both experiments, almost all KEGG terms show increased abundance 4.8, 4.9.

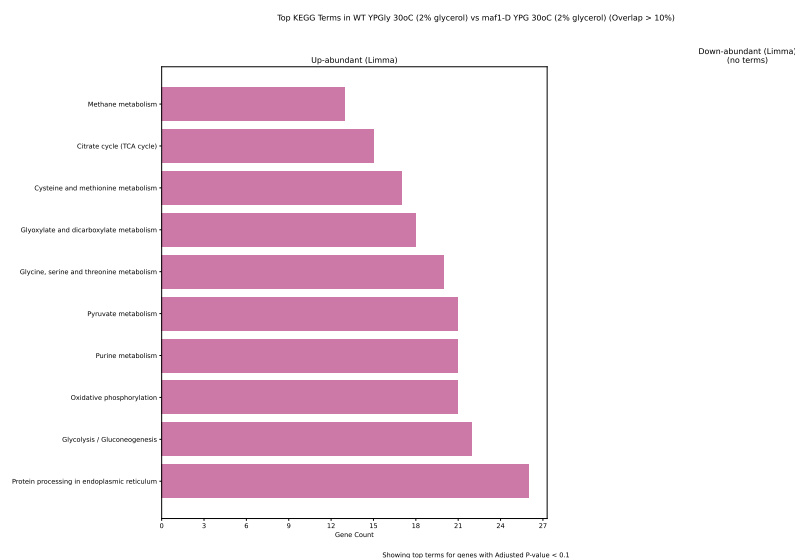


Figure 4.8: KEGG pathway enrichment for wild-type versus *maf1Δ* strains grown on glycerol at 30°C.

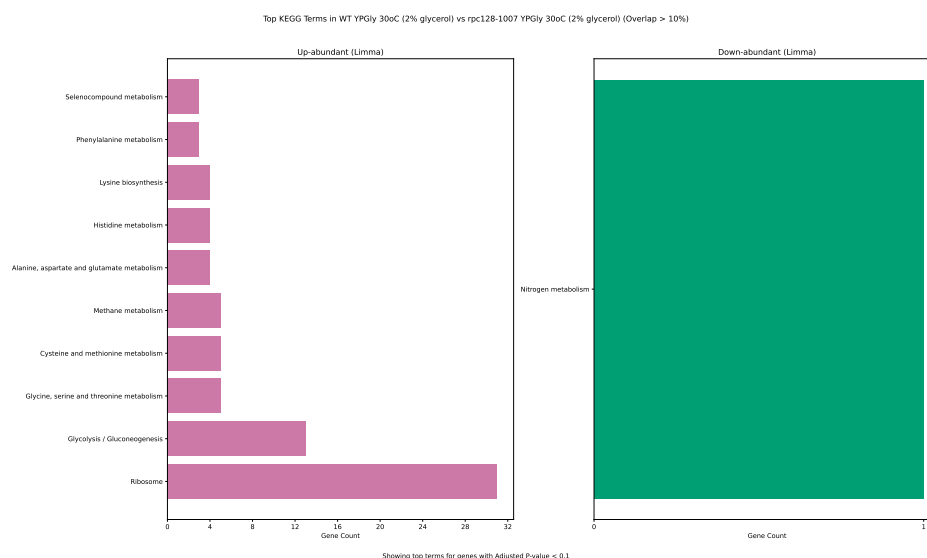


Figure 4.9: KEGG pathway enrichment for wild-type versus *rpc128-1007* strains grown on glycerol at 30°C.

Finally, elevating the temperature to 37°C for the *maf1Δ* strain on glycerol resulted in the downregulation of most selected pathways including pyrimidine metabolism, arginine biosynthesis (Figure 4.10). Phenylalanine metabolism was the only upregulated pathway in this condition. This broad downregulation of biosynthetic pathways aligns with the growth

arrest and lethality of the *maf1* Δ mutant under non-permissive conditions.

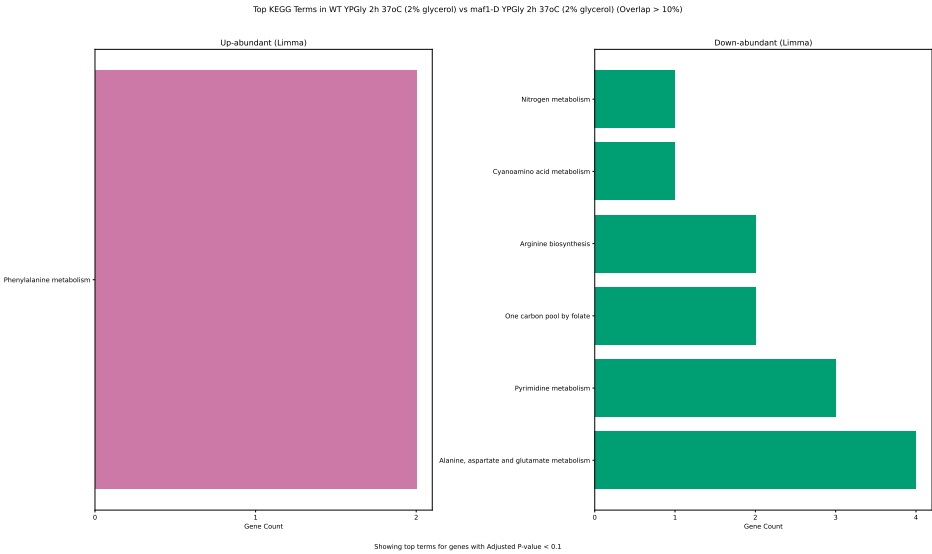
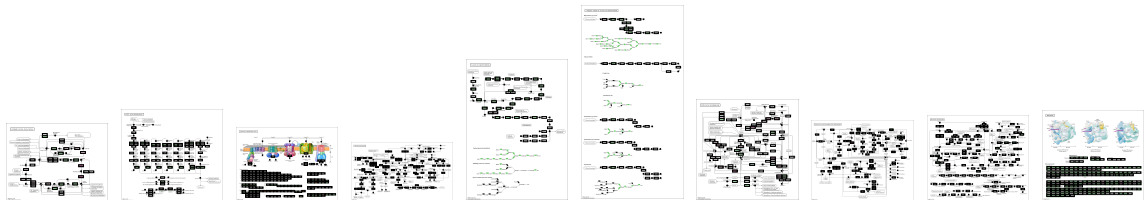


Figure 4.10: KEGG pathway enrichment for wild-type versus *maf1* Δ strains grown on glycerol at 37°C.

4.6. KEGG

Ścieżki do wybrania (na podstawie wcześniejszych wykresów) i opisanania.

A. *maf1*- Δ vs WT (YPD)



4.7. Functional Analysis

In the *maf1* Δ mutant, we found a significant enrichment of proteins related to cellular stress, particularly molecular chaperones (Figure 4.114.12). Finally, mapping the proteomic changes onto the KEGG [5] pathway for Oxidative Phosphorylation revealed widespread alterations in the abundance of OXPHOS complex subunits in the *maf1* Δ and *rpc128-1007* mutants (Figure 4.14).

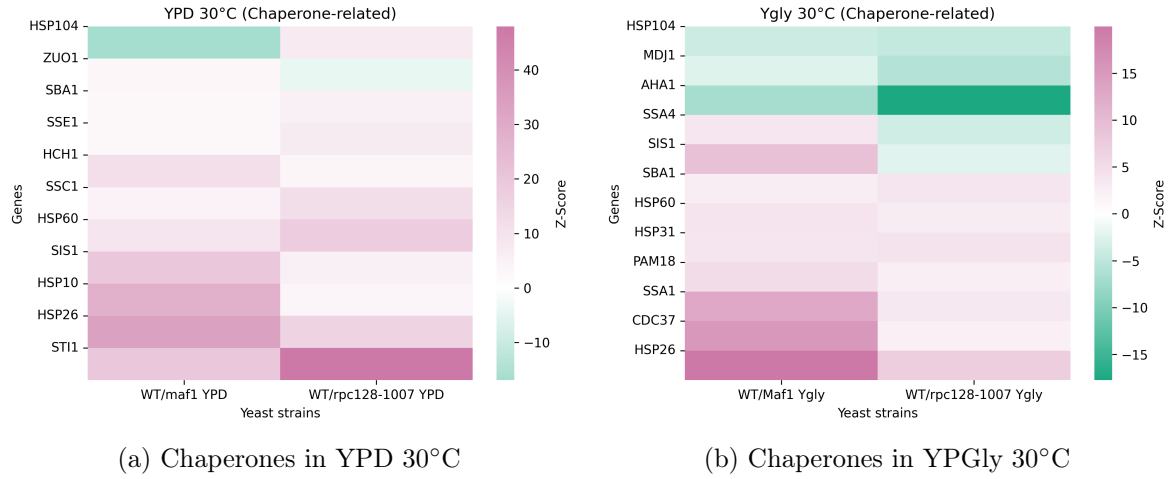


Figure 4.11: Expression of Chaperone-Related Proteins. Heatmaps showing the expression of chaperones and other stress-response proteins in different conditions.

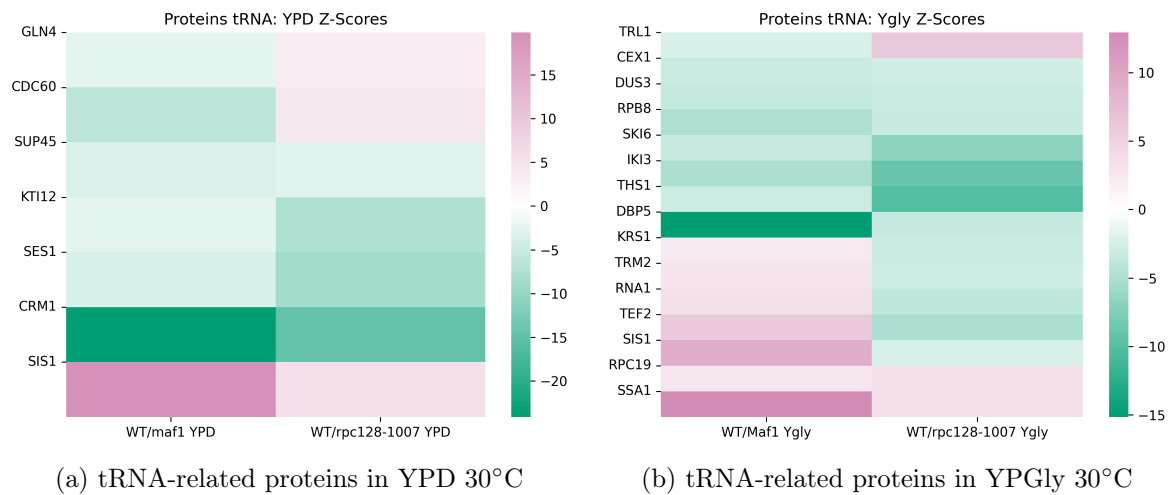


Figure 4.12: Expression of tRNA Metabolic Process Proteins. Heatmaps showing the differential expression of proteins involved in tRNA metabolism.

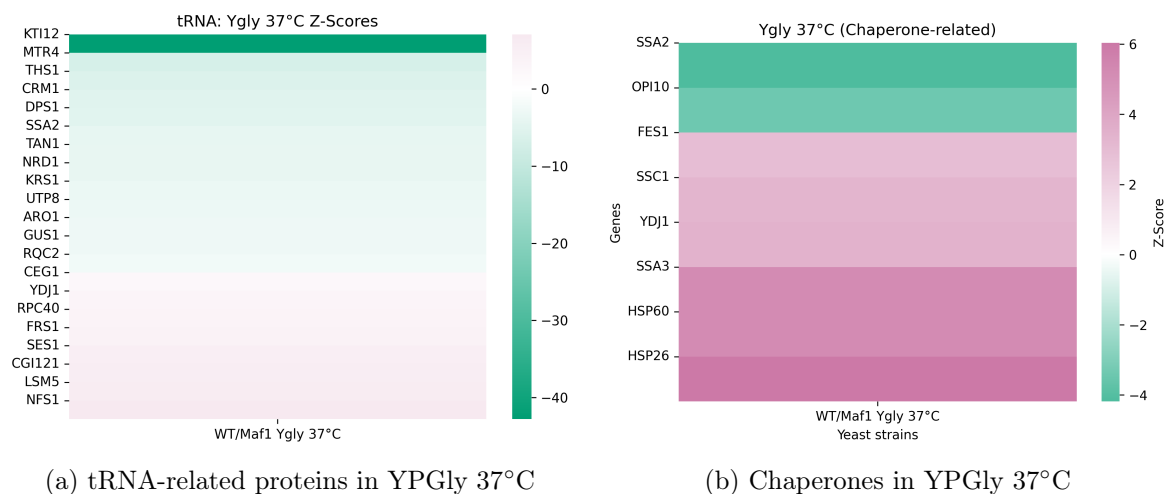


Figure 4.13: Expression of tRNA Metabolic Process Proteins. Heatmaps showing the differential expression of proteins involved in tRNA metabolism.

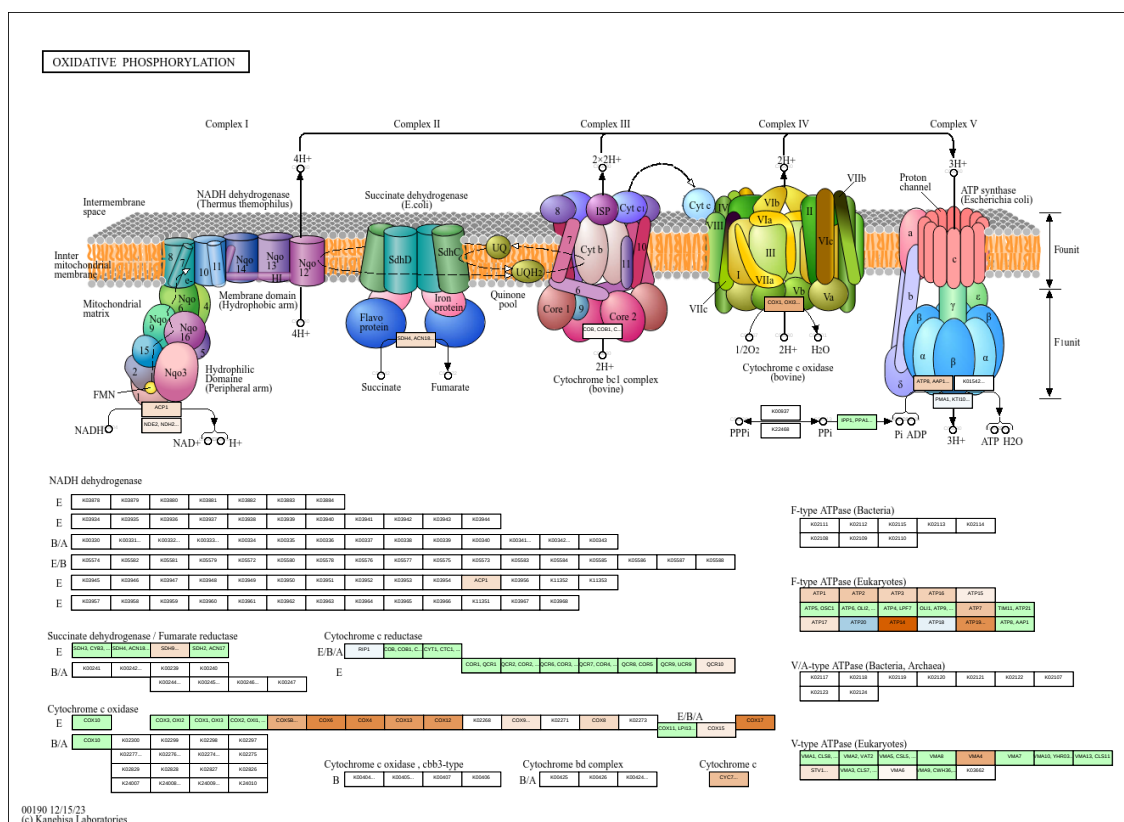


Figure 4.14: KEGG Pathway Map of Oxidative Phosphorylation. The colored boxes represent the log fold change of the corresponding protein subunits (KEGG Copyright Permission: 240975).

4.8. Integrative Analysis of Mutant and Laboratory Strains

To place the findings from the RNAP III mutants into a broader context, we performed an integrative analysis, combining our data with a published dataset from strains W303 and BY4742. The initial step revealed a strong batch effect, where samples clustered by their dataset of origin (Figure 4.15A, C). Applying the ComBat algorithm removed this technical variation (Figure 4.15B, D).

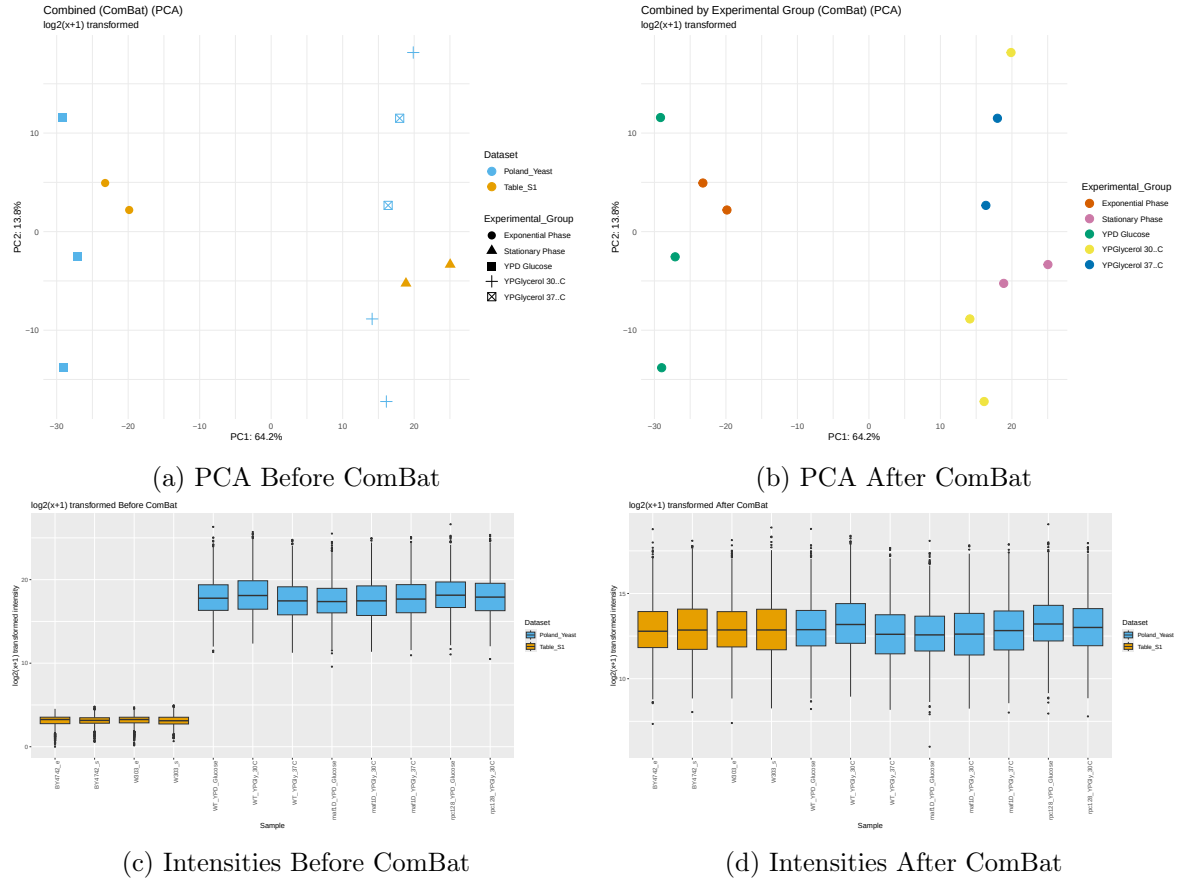


Figure 4.15: Batch Effect Correction with ComBat. (A, C) Before correction, PCA and boxplots show samples clustering by dataset, indicating a strong technical batch effect. (B, D) After applying ComBat, the non-biological variation is removed, and samples cluster by biological group, with normalized intensity distributions across datasets.

With the batch effect corrected, we explored proteomes of all strains. A heatmap of the combined data reveals distinct protein expression clusters (Figure 4.16). Dimensionality reduction with UMAP confirms similar clustering structure (Figure 4.17).

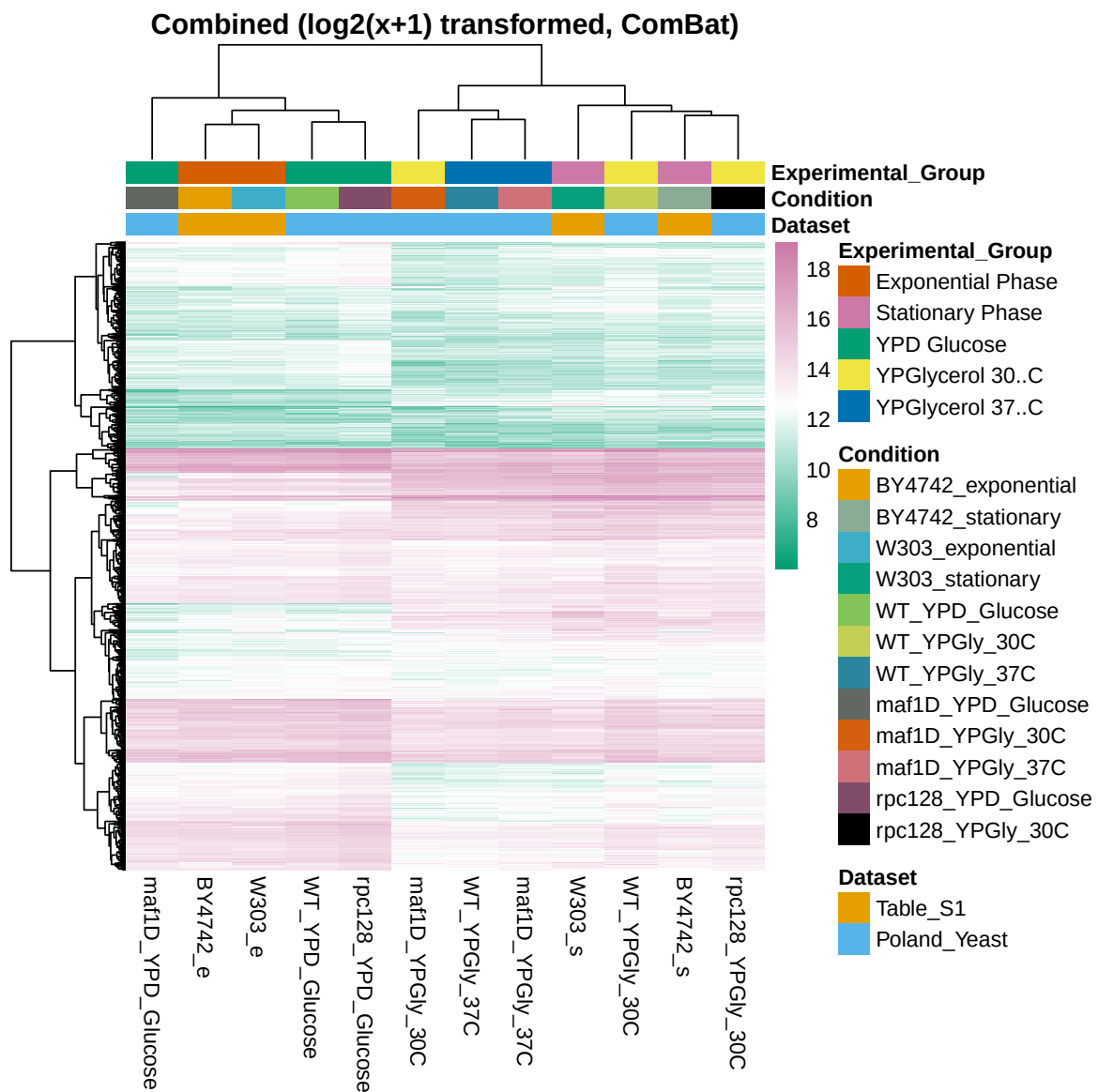


Figure 4.16: Heatmap of the Combined and Corrected Proteome. This heatmap displays the logarithmic scaled abundance of 1,589 common proteins across all samples from both experiments. Samples are annotated by experimental group, condition, and original dataset.

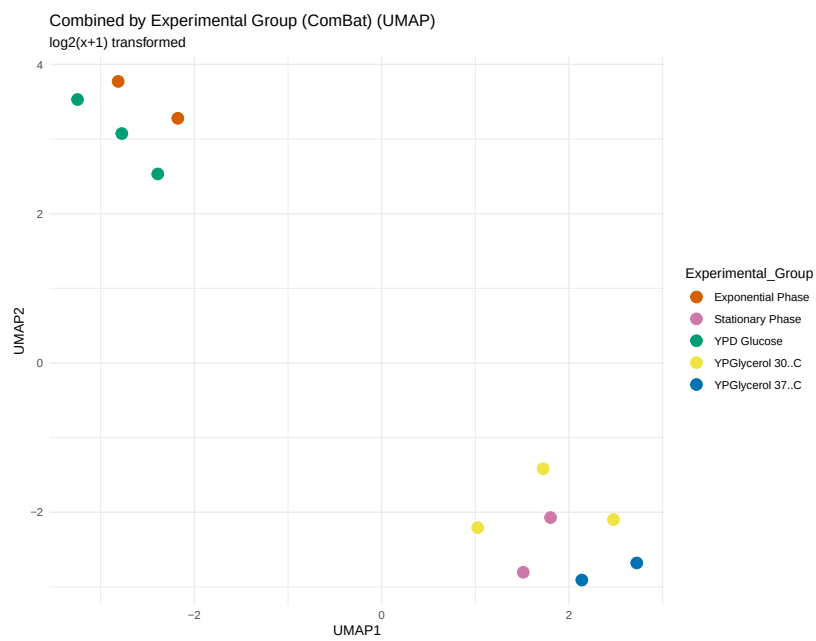


Figure 4.17: UMAP of the Combined and Corrected Proteome. The UMAP plot shows sample clustering based on their global proteomic profiles after batch correction.

Chapter 5

Additional Materials

Figure 5.1: Overview of the SWATH-MS workflow. Reproduced from [7]. This figure is licensed under the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>). No changes were made to the original image.

Bibliography

- [1] *Coefficient of Variation*, pages 95–96. Springer New York, New York, NY, 2008.
- [2] Vaibhav Gandhi. Chapter 2 - interfacing brain and machine. In Vaibhav Gandhi, editor, *Brain-Computer Interfacing for Assistive Robotics*, pages 7–63. Academic Press, San Diego, 2015.
- [3] Ludovic C. Gillet, Pedro Navarro, Stephen Tate, Hannes Röst, Nathalie Selevsek, Lukas Reiter, Ron Bonner, and Ruedi Aebersold. Targeted data extraction of the ms/ms spectra generated by data-independent acquisition: A new concept for consistent and accurate proteome analysis*. *Molecular & Cellular Proteomics*, 11(6):O111.016717, 2012.
- [4] W. Evan Johnson, Cheng Li, and Ariel Rabinovic. Adjusting batch effects in microarray expression data using empirical bayes methods. *Biostatistics*, 8(1):118–127, 04 2006.
- [5] Minoru Kanehisa, Miho Furumichi, Yoko Sato, Yuriko Matsuura, and Mari Ishiguro-Watanabe. Kegg: biological systems database as a model of the real world. *Nucleic Acids Research*, 53(D1):D672–D677, 10 2024.
- [6] Minoru Kanehisa and Susumu Goto. Kegg: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research*, 28(1):27–30, 01 2000.
- [7] Christina Ludwig, Ludovic Gillet, George Rosenberger, Sabine Amon, Ben C. Collins, and Ruedi Aebersold. Data-independent acquisition-based SWATH-MS for quantitative proteomics: a tutorial. *Molecular Systems Biology*, 14(8):e8126, 2018. Published under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). No changes were made.
- [8] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction, 2020.
- [9] an integrated chromatography MtoZ Biolabs and mass spectrometry (MS) services provider. Beginner’s guide to swath-ms proteomics: Principles, advantages, and applications. <https://www.mtoz-biolabs.com/beginners-guide-to-swath-ms-proteomics-principles-advantages-and-applications.html>. Accessed: 2026-03-05.
- [10] Marta Płonka, Donata Wawrzycka, Robert Wysocki, Magdalena Boguta, and Małgorzata Cieśla. Coupling of rna polymerase iii assembly to cell cycle progression in *saccharomyces cerevisiae*. *Cell Cycle*, 18(4):500–510, 2019. PMID: 30760101.
- [11] Valentina Rossio, Xinyue Liu, and Joao A. Paulo. Comparative proteomic analysis of two commonly used laboratory yeast strains: W303 and by4742. *Proteomes*, 11(4), 2023.

- [12] Roza Szatkowska, Emil Furmanek, Andrzej M. Kierzek, Christian Ludwig, and Malgorzata Adamczyk. Mitochondrial metabolism in the spotlight: Maintaining balanced rnap iii activity ensures cellular homeostasis. *International Journal of Molecular Sciences*, 24(19), 2023.
- [13] Qianmin Wang, Ming Lei, and Jian Wu. A structural perspective of human rna polymerase iii. *RNA Biology*, 19(1):246–255, 2022. PMID: 35133940.